

Early and late rate of force development: differential adaptive responses to resistance training?

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The objective of this study is to investigate the potentially opposing influence of qualitative and quantitative muscular adaptations in response to high-intensity resistance training on contractile rate of force development (RFD) in the early (<100 ms) and later phases (>200 ms) of rising muscle force. Fifteen healthy young males participated in a 14-week resistance training intervention for the lower body and 10 matched subjects participated as controls. Maximal muscle strength (MVC) and RFD were measured during maximal voluntary isometric contraction of the quadriceps femoris muscle. Muscle biopsies were obtained from the vastus lateralis. The main findings were that RFD in the late phase

of rising muscle force increased in response to resistance training whereas early RFD remained unchanged and early relative RFD (i.e., RFD/MVC) decreased. Quantitatively, muscle fiber cross-sectional area and MVC increased whereas, qualitatively, the relative proportion of type IIX muscle fibers decreased. Multiple regression analysis showed that while increased MVC positively influenced both early and late RFD, decreased-type IIX negatively influenced early RFD only. In conclusion, early and late RFD responded differently to high-intensity resistance training due to differential influences of qualitative and quantitative muscular adaptations on early and later phases of rising muscle force.

Explosive muscle strength is one of the single most important physiological parameters for successful performance in many sports. In the laboratory, a standardized method to assess explosive muscle strength *in vivo* is using the contractile rate of force development (RFD) during maximal isometric contraction (Bell & Jacobs, 1986; Aagaard et al., 2002). A similar approach has been applied for isolated animal muscles *in vitro* (Wilkie, 1949). While maximum muscle force is reached in the excess of 300 ms during a maximal voluntary contraction (Thorstensson et al., 1976), many sports allows for <250 ms to develop force, e.g. 110–160 ms in long jump (Luhtanen & Komi, 1979), 180–220 ms in high jump (Dapena & Chung, 1988) and 80–120 ms in sprint running (Kuitunen et al., 2002). Hence, contractile RFD rather than maximal muscle strength (MVC) becomes of paramount importance for performance in explosive-type sports.

Contractile RFD is a muscle mechanical parameter that is influenced by a multitude of physiological parameters (Aagaard, 2003), which include a relative proportion of slow and fast twitch muscle fibers (Harridge, 1996), muscle cross-sectional area (Aagaard, 2003), viscoelastic properties of the

muscle–tendon complex (Wilkie, 1949; Bojsen-Moller et al., 2005) and efferent neural drive to the muscle fibers (Grimby et al., 1981; Van Cutsem et al., 1998; Aagaard et al., 2002).

Using a cross-sectional study design, we recently demonstrated that contractile RFD in different time intervals from the onset of muscle contraction is affected by different physiological parameters (Andersen & Aagaard, 2006). While RFD recorded in the very early time interval was affected by intrinsic contractile properties of the muscle (defined as electrically evoked twitch RFD), RFD obtained in the later phases of rising muscle force was closely related to MVC (Andersen & Aagaard, 2006). It is well known that high-intensity resistance training increases MVC, however, at the expense of a decreased relative proportion of muscle fibers containing myosin heavy chain IIX (Andersen & Aagaard, 2000; Andersen et al., 2005) that are characterized by a high intrinsic RFD (Harridge, 2007). Taken together, this suggests that early- and late-phase RFD may change in different ways in response to resistance training. Consequently, analyzing RFD in the early and late phases of rising muscle force, respectively, seems of vital importance to elucidate

the potentially opposing effects of quantitative and qualitative muscular adaptations on RFD. This could have important consequences for optimal design of sports-specific resistance training. While increases in contractile RFD have been observed following different forms of resistance training (Hakkinen et al., 1985b; Schmidtbleicher & Buehrle, 1987; Young & Bilby, 1993; Van Cutsem et al., 1998; Aagaard et al., 2002; Suetta et al., 2004; Caserotti et al., 2008), this is not always a uniform finding (Wilson et al., 1993; Baker et al., 1994; Sleivert et al., 1995; Hakkinen et al., 1998). These contrasting results may be related to different training protocols, but may also be influenced by the method of determining RFD.

The aim of the present study was to analyze the change in RFD in incrementing time intervals of 0–10, 0–20, . . . , 0–250 ms from onset of muscle contraction induced by a period of high-intensity resistance training. We hypothesized that RFD obtained in the early (<100 ms) and later (>200 ms) phase of rising muscle force, respectively, would respond differently in response to high-intensity resistance training. Muscle fiber characteristics and MVC were determined in addition to the assessment of RFD to elucidate some of the underlying adaptation mechanisms.

Materials and methods

Study design

Fifteen healthy sedentary male subjects (23.5 ± 3.2 years, 179 ± 8 cm, 75.4 ± 9.6 kg, mean $\pm$ SD) participated in the resistance training intervention. Ten subjects matched for age, weight and height (23.9 ± 3.1 years, 181 ± 8 cm, 76.0 ± 5.0 kg) were recruited as a control group that did not engage in training. All participants were untrained in the sense that they had not previously participated in regular resistance training and none had participated in regular exercise within the last year. Resistance training was performed for 14 weeks with a total of 38 training sessions, corresponding to approximately 3 sessions/week. The resistance training program consisted of four to five sets of each of the following machine exercises: leg press, hack squat, knee extension and hamstring curl (Nordic Gym). Subjects were instructed to lift and lower the weight stack in a controlled manner to avoid injuries, i.e. performing the movements without sudden jerk or acceleration. The program was designed according to the linear progression model. During the first 15 sessions, the relative exercise loading was 10–12 RM (four sets), followed by 8–10 RM loads (four sets) during sessions 16–25 and heavier loads of 6–8 RM (five sets) during sessions 26–38. All training sessions were surveyed and supervised by the authors of the study. All subjects gave written informed consent to participate in the study, which was approved by the local Ethics Committee (KF-01-010/01).

Assessment of contractile RFD and MVC

Maximal voluntary isometric muscle contractions were performed for the knee extensors (m. quadriceps femoris) at a static knee joint angle of 70° (0° = full extension) using a KinCom dynamometer (Kinetics Communicator, Chattecx Corp., Chattanooga, Tennessee, USA) according to procedures described previously (Aagaard et al., 2002). Briefly, testing was

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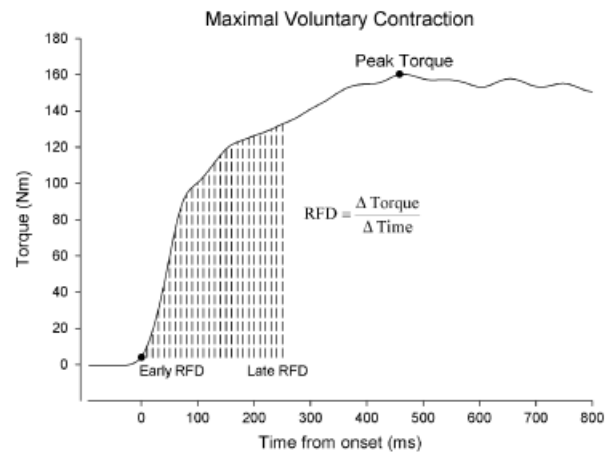


Fig. 1. Representative force (torque) tracing obtained during maximal voluntary quadriceps contraction in a single subject, showing peak torque as well as the time periods of 0–10, . . . , 0–250 ms from onset of contraction, which were identified for the determination of the rate of force development.

preceded by a 15 min warm up and pre-conditioning to the dynamometer during which a number of dynamic and isometric submaximal and near-maximal contractions were performed. Then, three maximal isometric attempts were performed with rest periods of 60 s in between. The subjects were instructed to extend their knee “as fast and as hard as possible” (Bemben et al., 1990; Sahaly et al., 2001) and each maximal contraction was sustained for approximately 3 s. Strong verbal encouragement was given during each trial. Online visual feedback of the dynamometer force signal was provided on a computer screen. Trials with an initial counter movement were discarded and an extra trial was performed (Grabiner, 1994). The dynamometer force signal was sampled at 1000 Hz (dt-9804, Data Translation) and stored in an external computer.

For later analysis, custom-made software was developed in MATLAB 6.5 (MathWorks, Natick, Massachusetts, USA) and VISUAL BASIC FOR APPLICATIONS (Microsoft Corp., Microsoft Denmark ApS, Hellerup, Denmark). The force signal was digitally lowpass filtered at 15 Hz using a fourth-order zero phase lag Butterworth filter (Winter, 1990) and subsequently multiplied by the lever arm length to obtain knee joint torque. The torque signal was corrected for the effect of gravity on the lower leg (Aagaard et al., 1995). MVC was defined as the highest peak torque value of the three maximal attempts. RFD (Nm/s) was defined as the slope of the torque–time curve (i.e., $\Delta \text{Torque} / \Delta \text{Time}$) in incrementing time periods of 0–10, 0–20, 0–30, . . . , 0–250 ms from the onset of contraction (Andersen & Aagaard, 2006). Onset of muscle contraction was defined as the instant when the knee extensor torque exceeded the baseline by 2.5% of the baseline-to-peak amplitude. For each time period, the highest value of the three maximal attempts was used for further analysis (Andersen & Aagaard, 2006). Relative RFD was calculated as RFD normalized to MVC (% MVC/s) in the above time periods. A representative torque–time curve is shown in Fig. 1.

Muscle biopsies

Muscle biopsies were obtained from subjects from the resistance training group from the middle part of their right vastus lateralis muscle (Bergström, 1962) within 1 week before the first training bout and 20–28 h after the last training bout. The biopsies were immediately mounted with Tissue-Tek, frozen in isopentane cooled with liquid nitrogen, and stored at -80°C

until further processing. Serial sections (10 μm) of the muscle biopsy samples were cut in a cryostat at -20°C and ATPase histochemistry analysis was performed after pre-incubation at pH of 4.37, 4.60 and 10.30 (Brooke & Kaiser, 1970). Immunocytochemistry was performed with two antibodies: BA-F8, reactive with MHC I and SC-71, reactive with MHC IIA and MHC IIX (Smerdu et al., 1994). On the basis of the ATPase and immunostaining, three different fiber types were defined (types I, IIA and IIX) (Andersen & Aagaard, 2000). The serial sections were visualized and analyzed using a light microscope connected to a camera combined with image analyzing software (Tema; Scanbeam, Hadsund, Denmark). Using the ATPase 4.6 staining, a fiber mask was drawn along the cell borders of the desired number of fibers. Images of the remaining ATPase stainings and the immunocytochemical stainings were then fitted into the fiber mask. A number was assigned by the computer to each specific fiber and the fibers were then displayed on the screen in multiple images. The individual fibers were therefore easily identified and assigned to a specific group of fiber type, allowing for determination of the relative proportion of the various fiber types, fiber type areas and fiber type sizes. On average, 269 ± 124 fibers (mean $\pm$ SD) were used to determine the fiber type distribution in each of the biopsies. Only truly horizontally cut fibers were used in the determination of the fiber size. Thus, a restricted number of fibers including at least 50 type I and 50 type II fibers were used for this analysis. Because the number of type IIX fibers in the post-training biopsies was highly restricted, all type IIA and IIX fibers were collapsed into type II fibers. On average, 153 ± 129 (mean $\pm$ SD) fibers were used for determining fiber size in each biopsy.

Statistical analysis

Changes in the main parameters were evaluated with analysis of variance using the mixed model procedure in the SAS VERSION 9.1 (SAS Institute, Cary, North Carolina, USA). Factors included in the model for RFD were Test Round (pre- and post-intervention), Contraction Time (0–10, 0–20, ..., 0–250 ms from onset of muscle contraction) and Group (resistance training and control group). Factors included in the model for muscle fiber size were Test Round and Fiber Type (type I and II). Factors included in the model for muscle fiber type percentage and area percentage were Test Round and Fiber Type (type I, IIA and IIX). When a significant main effect was found, *post hoc* tests were performed to locate differences. Pearson's correlation coefficient between changes in the main variables was calculated. Furthermore, multiple regression analysis was performed, with changes in RFD as dependent variable and changes in the other main variables as independent variables. An alpha level of 5% was accepted as statistically significant and all results were reported as mean SE unless otherwise stated.

Results

Training and compliance

Thirty-eight training sessions were planned during the 14-week intervention period. The actual number of training sessions was 35 ± 0.4 (range 34–38). The training load was increased markedly during the 14 weeks, which confirmed the high compliance documented in the logbooks of the participants. From the initial to the final week of training, the numerical load in the exercises increased from 97 ± 5 to 302 ± 18 kg for leg press, from 42 ± 3 to 123 ± 3 kg for knee

extension, from 54 ± 4 to 229 ± 16 kg for hack squat and from 29 ± 2 to 48 ± 1 kg for hamstring curls.

RFD

A priori hypothesis testing of main effects showed a significant Group by Contraction Time by Test Round effect for RFD ($F = 5.54$, $P < 0.001$) as well as relative RFD ($F = 24.4$, $P < 0.0001$).

Post hoc tests showed that in the resistance training group, the RFD increased 11% at 250 ms from onset of contraction ($P < 0.05$) whereas no significant changes occurred in the earlier time intervals (Fig. 2(a)). The training-induced increase in RFD observed at 250 ms was positively correlated to the gain in MVC ($r = 0.69$, $P < 0.01$) (Fig. 3(a)). Relative RFD (i.e., RFD/MVC) decreased 10–18% at time intervals up to 140 ms from onset of contraction ($P < 0.05$) whereas no change occurred in the later time intervals (Fig. 2(b)). The training-induced decrease in early relative RFD (e.g., at 50 ms) was positively correlated with the decrease in the area percentage of type IIX muscle fibers ($r = 0.61$, $P < 0.05$) (Fig. 3(b)). In the control group, RFD and relative RFD remained unchanged (Fig. 2(c) and (d)).

MVC

MVC increased 18% (from 258 ± 11 to 305 ± 15 Nm, $P < 0.001$) in the resistance training group whereas no change occurred in the control group (259 ± 16 vs 255 ± 13 Nm.).

Muscle fiber characteristics

The resistance training group demonstrated a significant Test Round by Fiber Type effect for muscle fiber cross-sectional area (CSA) ($P < 0.05$), muscle fiber percentage ($P < 0.001$) and muscle fiber area percentage ($P < 0.0001$).

Post hoc tests showed that type II muscle fiber CSA increased 23% (from 4714 ± 240 to $5782 \pm 371 \mu\text{m}^2$, $P < 0.01$) while remaining statistically unchanged for type I muscle fibers [3981 ± 223 vs $4324 \pm 221 \mu\text{m}^2$, not significant (NS)]. The relative proportion of type I, IIA and IIX fibers was unchanged (58 ± 2.5 vs $56 \pm 1.9\%$, NS), increased (from 31 ± 1.8 to $41 \pm 2.0\%$, $P < 0.001$) and decreased (from 11 ± 1.7 to $3.9 \pm 1.0\%$, $P < 0.05$), respectively. The area percentage tended to decrease for type I fibers (from 54 ± 2.5 to $49 \pm 2.0\%$, $P = 0.06$) while increasing (from 36 ± 2.1 to $47 \pm 2.2\%$, $P < 0.0001$) and decreasing (from 9.9 ± 1.6 to $4.0 \pm 1.1\%$, $P < 0.05$) for IIA and IIX fibers, respectively.

Multiple regression analysis

Changes in late-phase RFD (at 250 ms) were related to changes in MVC only, with an explained variance of 48% ($P < 0.05$). Changes in early-phase RFD (at

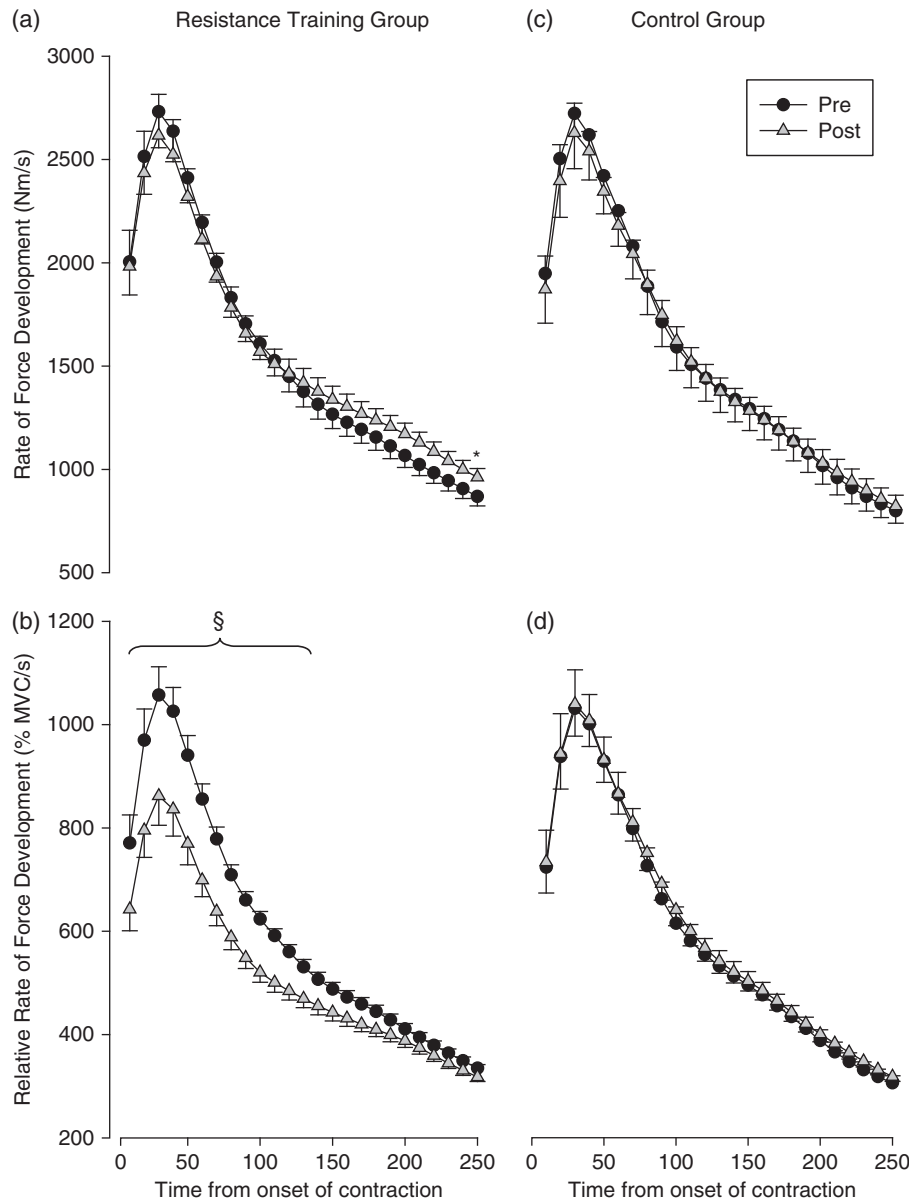


Fig. 2. (a–d) Contractile rate of force development (RFD) and relative RFD [i.e., RFD/maximal muscle strength (MVC)] in the resistance training group and control group at pre- and post-intervention. *RFD increased in the resistance training group at 250 ms from onset of contraction, $P < 0.05$. §Relative RFD decreased in the resistance training group in time periods up to 140 ms from onset of contraction, $P < 0.05$.

50 ms) were related to both changes in MVC and area percentage of type IIX, with a total explained variance of 58% ($P < 0.01$). Changes in early-phase relative RFD (at 50 ms) were related to changes in area percentage of type IIX only, with an explained variance of 38% ($P = 0.06$). The results of the multiple regression analysis are summarized in Table 1.

Discussion

The main finding of the present study was the differential changes in early and late RFD in response to 14 weeks of high-intensity resistance train-

ing. While RFD in the late and early phase of rising muscle force increased and remained unchanged, respectively, relative RFD in the early phase decreased. This was seen along with increased MVC, type II muscle fiber hypertrophy and decreased relative proportion of fast type IIX.

Adaptive gains in contractile RFD have previously been reported in response to different types of resistance training interventions, e.g. power training with relatively light loads and high movement speeds (Hakkinen et al., 1985b; Van Cutsem et al., 1998), resistance training involving high intensities of 1–12 RM and subsequently relatively slow movement speeds (Schmidtbleicher & Buehrle, 1987; Aagaard

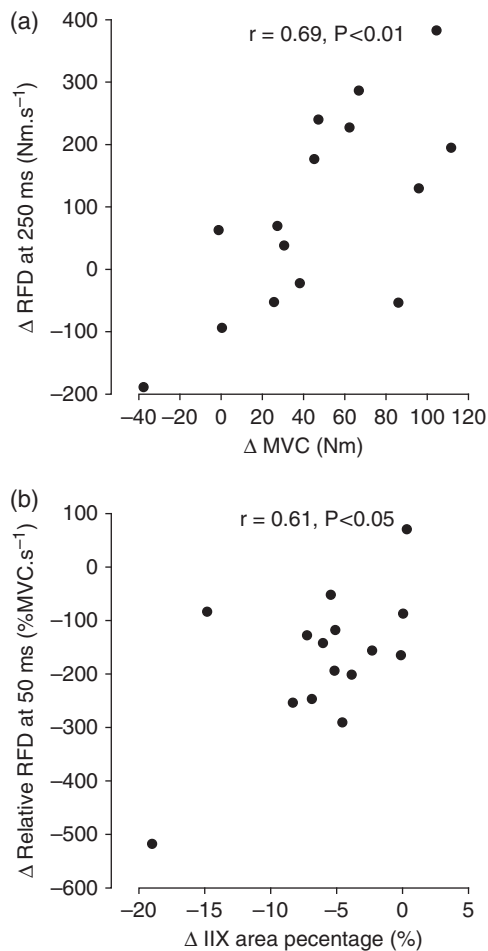


Fig. 3. Relationship (Pearson's r) between pre- to post-training changes in (a) maximal muscle strength (MVC) and late rate of force development (RFD) (250 ms) and (b) area percentage of type IIX and early relative RFD (50 ms) in the strength training group.

et al., 2002) and training with isometric muscle contractions executed with the intention of a high RFD (Behm & Sale, 1993). These training-induced adaptations are evident also in middle-aged and elderly individuals (Suetta et al., 2004; Barry et al., 2005; Caserotti et al., 2008). In contrast, yet other studies reported unchanged RFD in response to resistance training (Wilson et al., 1993; Baker et al., 1994; Sleivert et al., 1995; Hakkinen et al., 1998). The present study shows that the method of analyzing RFD is of paramount importance for the outcome and interpretation of results. While RFD at 250 ms increased in response to the present training intervention, relative RFD up to 140 ms from onset of contraction decreased.

We have previously shown in a cross-sectional study that contractile RFD obtained in the later phase of rising muscle force was closely related to the magnitude of MVC (Andersen & Aagaard, 2006). In the present study, we further observed that the training-induced rise in late-phase RFD was associated with the gain in MVC while not being nega-

Table 1. Multiple regression analysis for the pre- to post-training changes in the main variables

	Standardized parameter estimates (β)		R^2 (%)
	MVC	Area percentage type IIX	
Early RFD	0.62***	0.38*	58***
Late RFD	0.68***	0.05	48**
Early relative RFD	0.03	0.61**	38*
Late relative RFD	-0.05	-0.02	0

Changes in early- (50 ms) and late-phase RFD (250 ms) are dependent variables and changes in MVC and area percentage of type IIX are independent variables. Shown are the standardized parameter estimates (β 's) and the total explained variance of the model (R^2).

***, ***, * P < 0.10, 0.05 and 0.01, respectively.

MVC, maximal muscle strength; RFD, rate of force development.

tively influenced by decreased proportions of fast type IIX. In contrast, RFD obtained in the early phase of rising muscle force has been shown to be mostly related to the intrinsic contractile properties of the muscle and also to MVC (Andersen & Aagaard, 2006). The present study elaborates on these findings by showing that the explained variance in changes in early-phase RFD was related to changes in both MVC and area percentage of type IIX fibers. However, contrasting influences of these parameters, i.e. increased MVC and decreased proportion of fast type IIX muscle fibers, resulted in unchanged early RFD. The negative impact of decreased type IIX on early RFD became evident when RFD was normalized to MVC, i.e. decreased early relative RFD was related to decreased area percentage of type IIX fibers. Thus, the balance between increased muscle strength and decreased proportion of type IIX muscle fibers appears crucial for potential changes in early RFD.

In accordance with the present results, other research groups reported unchanged early-phase RFD in response to high-intensity resistance training (Hakkinen et al., 1985a, 1998; Narici et al., 1996). In contrast, we have previously observed elevated contractile RFD in both the early and the later phases of rising muscle force along with increased neuromuscular activity in response to high-intensity resistance training in both young (Aagaard et al., 2002) and aging individuals (Suetta et al., 2004). In the former study, very high intensities of 4 RM combined with tapering of total training volume was employed during the final weeks of training, which may explain the training-induced rise in early RFD. **In the present study, the subjects were instructed to lift and lower the weight in a controlled manner – and not with the intention of exerting a high RFD – which may partly explain that early RFD did not increase. Studies reporting gains in early-phase RFD typically have involved exercises that were executed with maximal intentional acceleration effort regardless of the actual movement velocity, i.e. with a high RFD (Hakkinen**

et al., 1985b; Schmidtbleicher & Buehrle, 1987; Behm & Sale, 1993; Van Cutsem et al., 1998; Suetta et al., 2004; Caserotti et al., 2008). Notably, Young and Bilby (1993) demonstrated that resistance training executed with highest possible acceleration increased RFD more than when the exercises were deliberately executed in a slow manner, in spite of comparable nominal loads lifted by the two intervention groups. Thus, when performing muscle contractions with a given load, the principal training stimulus for increasing early RFD may be that exercises are executed with the intention of maximal acceleration regardless of the actual movement velocity (Behm & Sale, 1993; Van Cutsem et al., 1998).

Whereas the present increase in late-phase RFD could be caused by quantitative gains in muscle fiber size and MVC, early-phase relative RFD could have decreased due to fiber type transition toward the slower type II phenotype, i.e. IIX $\rightarrow$ IIA. Multiple regression analyses confirmed that decreased type IIX could explain 38% of the decrease in early relative RFD. Maximal cross-bridge cycle transition rate appears to be a major limiting factor for the maximal intrinsic RFD of mammalian muscle fiber (Fitts & Widrick, 1996). Notably, cross-bridge cycling rate of type IIX muscle fibers markedly exceeds that of type I and IIA muscle fibers (Bottinelli et al., 1996; Harridge, 2007). Thus, adaptive changes in muscle fiber composition become especially important during the early phase of rising force where intrinsic muscle contractile properties dominate RFD (Andersen & Aagaard, 2006). Supporting this notion, a close association has previously been reported to exist between muscle fiber type composition and early RFD during electrically evoked maximal contraction (Harridge et al., 1996). Altogether, these findings suggest that decreased type IIX in response to resistance training was at least partially responsible for decreased relative RFD in the early phase of rising force.

Several other factors may potentially have affected the change in contractile RFD. Increases in maximal motor neuron firing frequency have previously been observed following resistance training (Van Cutsem et al., 1998; Kamen & Knight, 2004), which may contribute to an elevated RFD (Nelson, 1996; de Haan, 1998). Likewise, increased tendon stiffness has been reported following resistance training (Reeves et al., 2003; Kongsgaard et al., 2007), which may also *per se* lead to gains in RFD (Wilkie, 1949; Edman &

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Josephson, 2007). On the other hand, increased fascicle length and muscle fiber pennation angle in response to resistance training (Seynnes et al., 2007) may shift the joint torque–angle relationship. Potential changes in the torque–angle relationship after resistance training can have either a negative or a positive impact on contractile RFD, depending on the specific joint angle (i.e., muscle length) at which RFD is recorded (Blazevich et al., 2009).

In conclusion, the present study showed that contractile RFD in the early and later phases of rising muscle force responded differently in response to resistance training. While increased MVC positively influenced both early and late RFD, decreased type IIX negatively influenced early RFD only. Together, this resulted in increased late RFD, unchanged early RFD and decreased early relative RFD.

Perspectives

The present findings have important practical implications for the design of optimal training regimes as well as the interpretation of training-induced changes in RFD in relation to different types of sports. For instance, if the specific goal is to evaluate training-induced gains in explosive muscle strength in a long jumper, adaptations in RFD should be analyzed in the range of 110–160 ms from onset of muscle contraction (Luhtanen & Komi, 1979). The model of analysis put forward in the present study can be used to provide a detailed investigation of the adaptive responses in time-specific RFD evoked by different types of training and in different target groups. More studies should be undertaken to examine the effects of different types of resistance training on early- and late-phase RFD, respectively, to improve the design of optimal training regimes within sport and rehabilitation.

Key words: strength training, rate of torque development, fast twitch, slow twitch, ATPase, myosin heavy chain.

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